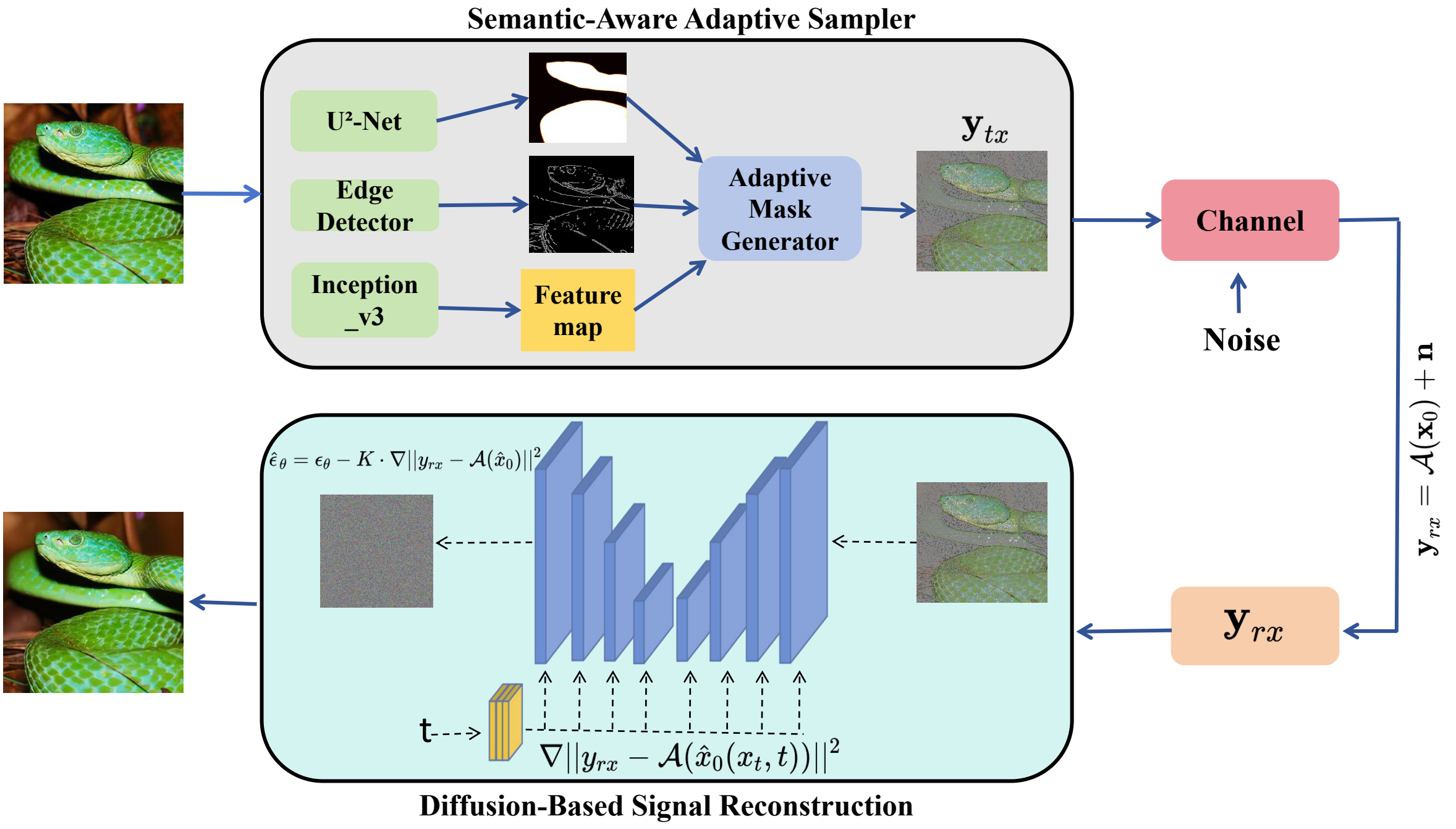
Semantic-Aware Adaptive Sampler
U²-Net
Edge Detector
Inception _v3
Feature map
Adaptive Mask Generator
$\mathbf{y}_{tx}$
Channel
Noise
$\mathbf{y}_{rx} = \mathcal{A}(\mathbf{x}_0) + \mathbf{n}$
$\hat{\epsilon}_\theta = \epsilon_\theta - K \cdot \nabla||y_{rx} - \mathcal{A}(\hat{x}_0)||^2$
t
$\nabla||y_{rx} - \mathcal{A}(\hat{x}_0(x_t, t))||^2$
$\mathbf{y}_{rx}$
Diffusion-Based Signal Reconstruction